Research Notes on Lasso regression

Lasso regression

$$L(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|_1$$

>> Section 1

That is, if regressors are uncorrelated, λ {\displaystyle \lambda } again specifies the minimal influence of β 0 {\displaystyle \beta _{0}} . Even when regressors are correlated, the first time that a regression parameter is activated occurs when λ {\displaystyle \lambda } is equal to the highest diagonal element of R ⊗ {\displaystyle R^{\otimes }} . These results can be compared to a rescaled version of the lasso by defining q lasso , i = 1 p ∑ l | b OLS , l − β 0 , l | {\displaystyle q_{{\mbox{lasso}},i}={\frac {1}{p}}\sum _{l}|b_{{\text{OLS}},l}-\beta _{0,l}|} , which is the average absolute deviation of b OLS {\displaystyle b_{\text{OLS}}} from β 0 {\displaystyle \beta _{0}} . Assuming that regressors are uncorrelated, then the moment of activation of the i t h {\displaystyle i^{th}} regressor is given by

>> Section 2

min β ∈ R p { 1 N ■ y − X β ■ 2 2 + λ ■ β ■ 1 } {\displaystyle \min _{\beta \in \mathbb {R} ^{p}}\left\{{\frac {1}{N}}\left\|y-X\beta \right\|_{2}^{2}+\lambda \|\beta \|_{1}\right\}}

>> Section 3

In statistics and machine learning, lasso (least absolute shrinkage and selection operator; also Lasso, LASSO or L1 regularization) is a regression analysis method that performs both variable selection and regularization in order to enhance the prediction accuracy and interpretability of the resulting statistical model. The lasso method assumes that the coefficients of the linear model are sparse, meaning that few of them are non-zero. It was originally introduced in geophysics, and later by Robert Tibshirani, who coined the term. Lasso was originally formulated for linear regression models. This simple case reveals a substantial amount about the estimator. These include its relationship to ridge regression and best subset selection and the connections between lasso coefficient estimates and so-called soft thresholding. It also reveals that (like standard linear regression) the coefficient estimates do not need to be unique if covariates are collinear. Though originally defined for linear regression, lasso regularization is easily extended to other statistical models including generalized linear models, generalized estimating equations, proportional hazards models, and M-estimators. Lasso's ability to perform subset selection relies on the form of the constraint and has a variety of interpretations including in terms of geometry, Bayesian statistics and convex analysis. The LASSO is closely related to basis pursuit denoising.

>> Section 4

j OLS | ≥ N λ] {\displaystyle \ {\hat {\beta }}_{j}\ =\ H_{\sqrt {N\lambda \ }}\ \left(\ {\hat {\beta }}{}_{j}^{\!\;{\text{OLS}}}\ \right)\ =\ {\hat {\beta }}{}_{j}^{\!\;{\text{OLS}}}\cdot \operatorname {\mathbb {I} } \left[~{\bigl |}{\hat {\beta }}{}_{j}^{\!\;{\text{OLS}}}{\bigr |}\geq {\sqrt {N\ \lambda \ }}~\right]\ }

>> Section 5

Generalizations Lasso variants have been created in order to remedy limitations of the original technique and to make the method more useful for particular problems. Almost all of these focus on respecting or exploiting dependencies among the covariates. Elastic net regularization adds an additional ridge regression-like penalty that improves performance when the number of predictors is larger than the sample size, allows the method to select strongly correlated variables together, and improves overall prediction accuracy. Group lasso allows groups of related covariates to be selected as a single unit, which can be useful in settings where it does not make sense to include some covariates without others. Further extensions of group lasso perform variable selection within individual groups (sparse group lasso) and allow overlap between groups (overlap group lasso). Fused lasso can account for the spatial or temporal characteristics of a problem, resulting in estimates that better match system structure. Lasso-regularized models can be fit using techniques including subgradient methods, least-angle regression (LARS), and proximal gradient methods. Determining the optimal value for the regularization parameter is an important part of ensuring that the model performs well; it is typically chosen using cross-validation.

>> Section 6

Just as ridge regression can be interpreted as linear regression for which the coefficients have been assigned normal prior distributions, lasso can be interpreted as linear regression for which the coefficients have Laplace prior distributions. The Laplace distribution is sharply peaked at zero (its first derivative is discontinuous at zero) and it concentrates its probability mass closer to zero than does the normal distribution. This provides an alternative explanation of why lasso tends to set some coefficients to zero, while ridge regression does not.

>> Section 7

j OLS | } {\displaystyle \,{\begin{aligned}{\hat {\beta }}_{j}\ =\ {}&\operatorname {S} _{N,\lambda }\left({\hat {\beta

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}}{}_{j}^{\!\;{\text{OLS}}}\right)\ =\ {\hat {\beta }}{}_{j}^{\!\;{\text{OLS}}}\cdot \max \!\left\{\ 0,\ 1-{\frac {\ N\ \lambda \ }{\ {\bigl |}{\hat {\beta }}{}_{j}^{\!\;{\text{OLS}}}{\bigr |}\ }}\ \right\}\end{aligned}}\,}

>> Section 8

but with respect to different constraints: ■ β ■ 1 ≤ t {\displaystyle \|\beta \|_{1}\leq t} for lasso and ■ β ■ 2 2 ≤ t {\displaystyle \|\beta \|_{2}^{2}\leq t} for ridge. The figure shows that the constraint region defined by the ■ 1 {\displaystyle \ell ^{1}} norm is a square rotated so that its corners lie on the axes (in general a cross-polytope), while the region defined by the ■ 2 {\displaystyle \ell ^{2}} norm is a circle (in general an n-sphere), which is rotationally invariant and, therefore, has no corners. As seen in the figure, a convex object that lies tangent to the boundary, such as the line shown, is likely to encounter a corner (or a higher-dimensional equivalent) of a hypercube, for which some components of β {\displaystyle \beta } are identically zero, while in the case of an n-sphere, the points on the boundary for which some of the components of β {\displaystyle \beta } are zero are not distinguished from the others and the convex object is no more likely to contact a point at which some components of β {\displaystyle \beta } are zero than one for which none of them are.

>> Section 9

For p = 1 {\displaystyle p=1} , the moment of activation is again given by λ ~ lasso , i = R 2 {\displaystyle {\tilde {\lambda }}_{{\text{lasso}},i}=R^{2}} . If β 0 {\displaystyle \beta _{0}} is a vector of zeros and a subset of p B {\displaystyle p_{B}} relevant parameters are equally responsible for a perfect fit of R 2 = 1 {\displaystyle R^{2}=1} , then this subset is activated at a λ {\displaystyle \lambda } value of 1 p {\displaystyle {\frac {1}{p}}} . The moment of activation of a relevant regressor then equals 1 p 1 p B p B 1 p B = 1 p {\displaystyle {\frac {1}{p}}{\frac {1}{\sqrt {p_{B}}}}p_{B}{\frac {1}{\sqrt {p_{B}}}}={\frac {1}{p}}} . In other words, the inclusion of irrelevant regressors delays the moment that relevant regressors are activated by this rescaled lasso. The adaptive lasso and the lasso are special cases of a '1ASTc' estimator. The latter only groups parameters together if the absolute correlation among regressors is larger than a user-specified value.